

KILLARI DHANUNJAY

M.Tech in Power Electronics and Drives
National Institute Of Technology, Warangal

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PROFESSIONAL SUMMARY

Power Electronics Engineer (M.Tech NIT Warangal, 8.18 CGPA) with hands-on hardware experience in DC-DC converters and inverter-fed PMSM drives using TI C2000 DSP. Expertise in closed-loop control, SVPWM, FOC, FCS-MPC, MATLAB/Simulink and Embedded C. GATE EE 3x qualified with consistent academic record.

EDUCATION

• National Institute of Technology, Warangal <i>M.Tech in Power Electronics and Drives</i>	<i>2025–Present</i> CGPA: 8.18
• Sagi Ramakrishnam Raju Engineering College, Bhimavaram <i>B.Tech in Electrical and Electronics Engineering</i>	<i>2020–2024</i> CGPA: 8.11
• Sri Aditya Junior College, Narsapuram <i>Board of Intermediate Education, Andhra Pradesh</i>	<i>2018–2020</i> CGPA: 9.7
• Sri Gowthami School, Narsapuram <i>Board of Secondary Education, Andhra Pradesh</i>	<i>2018</i> CGPA: 9.8

RESEARCH & HARDWARE EXPERIENCE

• National Institute of Technology, Warangal <i>Three-Phase Two-Level Inverter with DSP-Based SVPWM MATLAB/Simulink, TI C2000</i>	<i>Under Prof. Sharmili Das</i> May 2026 – July 2026
– Designed 3-phase 2-level VSI, 400V DC-link, 10kHz switching, SVPWM implementation. – Developed real-time PWM generation and dead-time insertion using TI C2000 ePWM modules. – Analyzed THD, modulation index vs V/f, and harmonic performance under varying load conditions. – Tools: MATLAB/Simulink, TI F28379D, Embedded C, Code Composer Studio	

ACADEMIC PROJECTS

• Advanced Control of Inverter-Fed PMSM Drive Using FOC, MPC and Hysteresis Control <i>3HP, 4-Pole, 1500RPM PMSM, 2-Level VSI with SVPWM MATLAB/Simulink, dSPACE DS1104</i>	<i>Feb 2026 – Apr 2026</i> NIT-Warangal
– Derived dq-axis model and implemented FOC with Clarke/Park transformations. – Developed FCS-MPC in Simulink for current regulation and compared with FOC-PI. – Implemented direct abc current hysteresis control without dq transformation on dSPACE. – Validated 3HP, 4-pole, 1500RPM PMSM drive with current sensors and encoder using ControlDesk. – Tools: MATLAB/Simulink, dSPACE, FOC, FCS-MPC, Hysteresis Control	
• Digital Control of Boost Converter Using TI TMS320F28379D <i>48V to 100V, 100W, 10kHz 100Ω Load Closed-loop Hardware Prototype</i>	<i>Jan 2026 – Mar 2026</i>
– Designed 48V to 100V, 100W Boost converter: L=6mH, C=220uF, R=100Ω, fsw=10kHz, D=0.52, MOSFET IRF540N, Gate Driver TLP250, Voltage Sensor LV25P. – Implemented voltage-mode PI control on TMS320F28379D with LV25P-based voltage sensing, ADC sampling and 10kHz ePWM in Embedded C (CCS). – Achieved 100V regulated output with <3% steady-state error, validated line/load transient response on hardware.	

TECHNICAL SKILLS

Power Electronics : DC-DC Converters, Inverters, PWM, SVPWM, Gate Drivers, SiC/GaN Devices.
Developer Tools : d-Space, MATLAB/Simulink, LTspice, PLECS, KICAD, ANSYS Maxwell.
Embedded Systems : TI C2000 F28379D, ADC, ePWM, PI Control, Motor Control Algorithms, CCS.
Programming Languages : C, Embedded C, Python (Basics).
Coursework : Power Electronics, Electric Drives, DSP (C2000), Wide Bandgap Devices, Micro-Grids, Control Systems.
Areas of Interest : Power Electronics, Motor Drives, EV Power Systems, Embedded Control.

ACHIEVEMENTS

• GATE 2024, 2025, 2026 Qualified 3 times - Electrical Engineering	<i>2024–2026</i>
• First Prize Project Expo, SRKR Engineering College	<i>2024</i>